

CROSSVILLE 7 NW, TN, USA

WMO: 754330

Lat: 36.0138N

Lon: 85.1345W

Elev: 1913

StdP: 13.71

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63855

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	14.6	1.5	6.3	12.2	6.1	8.1	17.0	13.2	43.5	12.5	42.2	3.4	N/A	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	86.8	72.1	84.7	72.1	83.0	71.7	75.3	81.6	74.3	80.7	73.3	79.7	4.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	133.2	78.5	72.3	128.6	77.5	71.3	124.1	76.6	40.1	81.8	39.2	80.9	38.3	80.0	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	10.5	8.9	2.6	89.7	6.2	4.3	-1.9	92.9	-5.5	95.4	-9.0	97.8	-13.5	100.9
			1.4	77.2	7.3	1.3	-3.9	78.1	-8.1	78.9	-12.2	79.7	-17.5	80.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	34.9	38.5	46.2	55.3	62.7	69.9	72.5	72.2	66.5	55.4	45.1	39.2
	DBStd	15.67	11.98	11.04	10.20	8.53	7.00	4.10	3.74	3.90	5.78	8.61	8.90	10.24
	HDD50	1680	485	345	197	46	6	0	0	0	0	46	195	360
	HDD65	4461	935	743	585	300	122	9	3	2	52	312	598	800
	CDD50	3485	15	22	78	206	401	597	699	687	495	213	47	25
	CDD65	791	0	0	1	9	52	157	237	224	97	14	0	0
	CDH74	5744	0	0	6	74	383	1143	1655	1630	725	125	3	0
	CDH80	1260	0	0	0	1	27	228	392	446	147	19	0	0
(14) Wind	WSAvg	3.7	4.9	5.0	5.1	4.7	3.3	2.7	2.6	2.4	2.4	3.1	3.9	4.6
(15) Precipitation	PrecAvg	59.8	5.4	5.0	6.2	5.0	5.6	4.8	5.3	4.0	4.1	3.5	5.0	5.9
	PrecMax	75.9	11.0	10.7	16.5	11.3	15.0	13.1	13.3	8.3	8.3	8.7	9.2	16.7
	PrecMin	42.7	1.3	1.1	1.3	1.5	1.9	0.9	1.1	1.5	0.4	0.0	1.8	0.8
	PrecStd	8.5	2.3	2.2	3.2	2.3	2.7	2.7	2.3	1.6	2.0	2.0	1.9	3.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.5	67.7	74.6	79.2	83.0	87.1	89.4	91.7	87.1	82.4	71.4	66.3
		MCWB	58.6	60.3	60.5	62.4	69.2	68.8	71.4	78.1	70.4	68.1	60.1	61.5
	2%	DB	59.5	63.6	70.5	76.2	80.4	84.4	86.0	87.3	83.9	77.2	66.8	62.4
		MCWB	54.6	59.3	58.2	60.7	67.4	71.7	73.5	72.5	69.8	64.4	55.9	57.7
	5%	DB	57.0	59.8	66.5	73.6	78.3	82.6	84.1	84.6	81.5	74.2	63.3	59.2
		MCWB	53.0	54.5	56.3	60.1	66.3	71.6	73.2	72.4	69.5	63.0	55.8	56.0
	10%	DB	53.7	55.9	62.6	70.7	76.0	80.5	82.1	82.2	78.4	70.3	60.3	55.4
		MCWB	50.2	50.8	54.3	58.9	65.3	70.7	72.7	71.7	68.8	61.9	54.7	51.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.0	62.6	62.2	67.1	71.4	75.3	76.9	76.8	73.4	71.2	64.3	63.5
		MCDB	62.0	65.9	69.9	73.8	79.7	82.4	82.7	83.9	80.5	77.2	67.3	65.6
	2%	WB	56.6	60.7	60.4	64.6	69.7	74.2	75.4	75.1	72.4	69.0	61.6	59.8
		MCDB	58.2	63.7	66.3	72.4	76.9	81.4	81.7	81.1	79.2	74.3	64.0	61.5
	5%	WB	53.7	57.7	58.7	62.9	68.5	73.0	74.6	73.9	71.2	66.1	59.1	56.3
		MCDB	55.5	60.8	64.2	69.9	75.5	80.0	80.9	80.0	77.8	70.8	61.0	58.7
	10%	WB	49.7	53.4	56.1	61.1	67.3	71.8	73.6	72.8	69.8	63.4	54.9	52.8
		MCDB	51.5	55.7	61.2	67.8	74.2	77.8	79.9	78.6	76.0	67.8	58.3	55.2
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	17.5	19.1	20.9	19.9	19.2	17.6	18.3	19.8	21.5	20.5	16.6
		MCDBR	17.5	20.1	23.3	24.0	22.6	21.7	20.1	22.0	23.7	24.9	23.4	17.8
		MCWBR	14.1	13.6	14.0	12.4	10.6	10.2	8.5	9.5	10.4	12.8	13.7	13.0
	5% WB	MCDBR	15.1	15.9	18.7	18.9	18.7	18.0	17.0	17.1	18.8	19.7	14.1	14.9
		MCWBR	13.7	12.9	12.9	11.2	9.8	9.8	8.5	8.8	9.3	12.2	11.7	13.1
(40) Clear-Sky Solar Irradiance	taub		0.299	0.311	0.338	0.374	0.419	0.449	0.478	0.486	0.393	0.353	0.316	0.296
	taud		2.525	2.482	2.441	2.329	2.229	2.172	2.120	2.083	2.328	2.441	2.508	2.549
	Ebn at Noon		287	296	295	288	275	266	257	252	273	276	278	281
	Edh at Noon		25	30	34	40	45	47	49	50	37	30	25	23
(44) All-Sky Solar Radiation	RadAvg		701	880	1244	1623	1797	1947	1842	1725	1460	1155	831	605
	RadStd		62	113	122	115	144	126	104	88	140	149	92	58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	+166

Nomenclature: See separate page